DBCM — Declared Behavior Comparison Module

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Canonical Definition

Declared Behavior Comparison Module defines the structural conditions under which a system's declared behavior and its actual live behavior may be compared, differentiated, reviewed, and judged for operational alignment so that behavior appearing compliant in description cannot silently diverge from behavior occurring in reality.

A system satisfies DBCM only if:

- declared behavior is explicitly defined
- actual behavior remains reviewable during or after runtime
- comparison logic between declared and actual behavior is structurally defined
- material divergence can be identified before it is normalized as acceptable operation
- the system does not continue claiming behavioral compliance while behaving differently in practice

A system that cannot compare what it says it does with what it actually does does not satisfy DBCM.

Module Function

DBCM defines the behavior-comparison layer of operational reality governance.

It ensures that operational validity is not assumed from what a system was expected, intended, documented, or configured to do, but remains tied to what the system actually does under live conditions.

The module applies wherever a system must compare:

- declared operating behavior
- actual runtime behavior
- expected control response
- actual action path
- documented workflow logic

- live system conduct
- stated output logic
- real consequence-producing behavior

Its function is not merely to detect activity.

Its function is to test whether behavior in reality remains the same behavior that was declared.

Step 1 — Define the Declared Behavior Object

The organization must define what behavior the system claims it performs.

Minimum requirement:

- the declared behavior object is explicit
- the behavioral claim is structurally bounded
- undefined or shifting behavioral claims are excluded from valid comparison logic

The declared behavior object may include:

- response behavior
- execution behavior
- decision behavior
- routing behavior
- intervention behavior
- permission-limited action behavior
- monitoring behavior
- escalation behavior

Without a clearly declared behavior object, comparison cannot begin.

Step 2 — Define Actual Behavior Observation Conditions

The system must define how actual live behavior is identified and observed.

Minimum requirement:

- actual behavior observation is explicit
- live behavior is not inferred only from intended configuration
- the system can distinguish performed behavior from declared behavior

This includes identifying:

- actual actions taken
- actual sequence of execution
- real intervention points
- actual routing logic
- live response conditions

- consequence-bearing behavior paths
- effective authority behavior in runtime

Without observable actual behavior, declared comparison becomes symbolic.

Step 3 — Define Comparison Logic

The system must define how declared behavior and actual behavior are compared.

Minimum requirement:

- comparison logic is explicit
- similarity of labels is not treated as sufficient behavioral alignment
- the system can distinguish full alignment, partial alignment, material divergence, and invalid behavioral mismatch

Comparison may evaluate:

- action type
- execution order
- timing
- scope
- authority condition
- intervention threshold
- escalation path
- outcome relevance

A system is not behaviorally aligned because it remains broadly similar in appearance.

It is aligned only when actual behavior materially matches declared behavior under governed comparison criteria.

Step 4 — Define Material Divergence Conditions

The system must define when behavioral divergence becomes operationally significant.

Minimum requirement:

- material divergence conditions are explicit
- not every variation is treated as invalid, but consequential divergence is not ignored
- the system can determine when actual behavior no longer supports the declared operational claim

Material divergence may include:

- performing actions not declared
- omitting declared controls in practice
- escalating behavior without declared basis
- routing behavior differently from stated logic

- changing intervention behavior under live conditions
- generating outputs under undeclared operational paths
- producing consequences inconsistent with declared behavioral design

Without material divergence logic, behavior drift becomes invisible
until failure is already normal.

Step 5 — Define Behavioral Continuity Conditions

The system must define how declared and actual behavior remain
aligned across runtime evolution.

Minimum requirement:

- behavioral continuity is explicit
- alignment is not assumed permanently after initial declaration
- changes in environment, runtime mode, delegation, escalation, or autonomy can affect behavioral alignment

This means the system must determine whether declared-versus-actual
alignment remains preserved during:

- updates
- runtime adaptation
- orchestration change
- permission change
- supervision change
- cross-system interaction
- escalating operational conditions

Step 6 — Preserve Comparison Traceability

The system must preserve traceability of declared behavior, actual
behavior, and comparison findings.

Minimum requirement:

- comparison events are reviewable
- divergence findings remain reconstructable
- later audit can determine what behavior was declared, what behavior occurred, how they were compared, and
- where alignment failed or remained intact

If behavioral comparison cannot be reconstructed, operational
reality becomes vulnerable to retrospective narrative rather than
structural review.

Step 7 — Restrict Invalid Behavioral Claim Continuity

The system must not be treated as valid if it continues to present
declared behavior as operationally true after material divergence
has already occurred.

Minimum requirement:

- invalid comparison conditions are identifiable
- behavioral claims unsupported by actual behavior are excluded
- systems operating under materially divergent behavior are blocked, narrowed, flagged, or invalidated where
- operational reality requires declared-versus-actual consistency

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Scenario

An AI-supported workflow declares that outputs are reviewed, bounded, or escalated through human oversight before consequence-bearing action occurs.

Application

DBCM is used to ensure:

- the declared review behavior is explicitly defined
- actual runtime workflow behavior is observable
- the system can compare declared human review logic with actual live behavior
- undeclared bypass of review does not remain hidden behind documented process language

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Result

The organization gains a stronger operational reality architecture in which declared review behavior must remain behaviorally real, not only procedurally stated.

Scenario

A system declares that it acts only within narrow control conditions, but under live runtime it begins taking actions, making routing decisions, or escalating behavior beyond what was originally declared.

Application

DBCM is used to ensure:

- declared behavioral scope is bounded
- actual live behavior remains observable
- divergence between declared and actual conduct can be identified
- the system cannot preserve a false behavioral identity while acting differently in practice

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Result

The environment gains a stronger comparison architecture in which operational behavior remains judged against reality rather than against outdated description.

Canonical Closing Statement

If declared behavior and actual behavior cannot be compared under governed conditions, operational reality collapses into descriptive fiction rather than live truth.

Related Documents

→ Operational Reality Standard (ORS™)

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Canonical Source

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